Q1. The photographs below show pupils investigating the movement of objects on ramps.



Plan an investigation into the factors affecting the movement of objects on ramps.

You can use any objects and any surfaces you like, and any other equipment you need.

In the box below, write a short draft of **one** question you could plan to investigate about the movement of objects on ramps.

Use your draft to help you answer the following questions.

(a) Give **one** factor you could change as you carry out your investigation (the independent variable).

.....

.....

1 mark

(b) What factor would you observe or measure to collect your results (the dependent variable) and what equipment would you use to measure them?

The factor I would observe or measure is .....

.....

1 mark

The measuring equipment I would use is .....

.....

1 mark

(c) Give **one** factor you should keep the same to make your test fair.

.....

.....

1 mark

maximum 4 marks

Q1.

(a) any suitable independent variable such as

- the surface
- the angle of the slope
- the kind of object
- the size of the push

*accept specific variations in objects, such as, 'weight' **or** 'mass' **or** 'surface area' **or** 'type of trainer sole' **or** 'type of shoe'*

1 (L4)

(b) any suitable dependent variable such as

- the distance travelled
- the time to move down the ramp
- the force needed to start the object moving
- the angle of the ramp at which the object starts moving

accept 'the time to reach a given point'
*accept 'angle **or** height of ramp'*
accept 'speed'
a dependent variable (DV) without an independent variable (IV) can gain credit

1 (L4)

any appropriate equipment to measure the dependent variable such as

- ruler **or** metre rule
- stopwatch **or** timer **or** light gates
- newton meter
- protractor

accept 'tape measure'
accept 'clock'
*do **not** accept a measurement strategy if a DV is not given **or** is incorrect*

1 (L4)

(c) any appropriate control variable such as

- the object used
- the angle of the slope

- the surface used
- the height of the ramp
- the length of the ramp

accept 'distance travelled'

only give credit for a control variable which does not conflict with the suggested investigation

1 (L4)

[4]